F FWallSim · Firewall Course

CLASS 11

SSL-VPN for remote workers

SSL-VPN settings, portals, ssl.root policy

Advanced | ~25 minutes | FWallSim v1.0.0 (FortiOS 7.4 style)

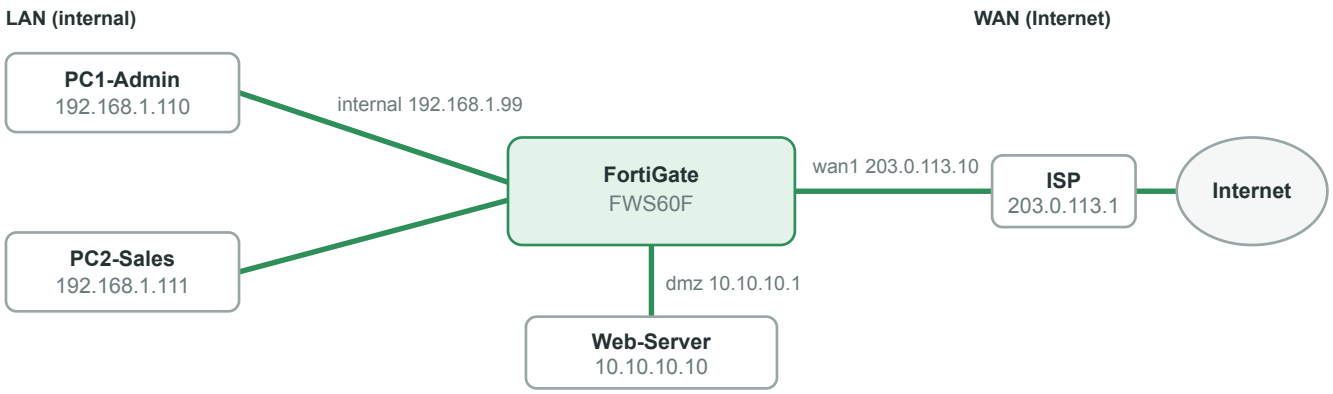
YOU WILL LEARN

- SSL-VPN settings
- portals
- ssl.root policy

BEFORE YOU START

Finish Class 10 first.

The lab network



The goal

THE IDEA

Remote staff need FortiClient access to the office PCs. Create a local user, a group, SSL-VPN settings on wan1 and the ssl.root policy, then connect Remote-Laptop.

How to start

1. Open FWallSim and log in.
2. Click **Labs** in the header.
3. Find **SSL-VPN for remote workers** and click **Start**.
4. The Lab Guide panel opens. Each task below gets a green tick as soon as it works.

Tasks

1. **SSL-VPN listens on wan1 on a port other than the admin HTTPS port**

   VPN > SSL-VPN Settings: Listen on wan1, port 10443.

2. **A local user is in a user group**

   User & Authentication > User Definition > Create New (Local User, with a password). Then User Groups > Create New (Firewall) and add the user as a member.

3. **An authentication rule maps that group to a tunnel-mode portal**

   VPN > SSL-VPN Settings > Authentication/Portal Mapping > Create New: your group, portal tunnel-access. That portal uses split tunneling, so only office traffic goes through the VPN and the laptop reaches the internet directly.

4. **A policy ssl.root → internal allows that group**

   Firewall Policy > Create New: Incoming SSL-VPN tunnel interface (ssl.root), Outgoing internal, Source all + your group, Destination the office LAN, Service ALL, ACCEPT.

5. **Remote-Laptop is connected with FortiClient**

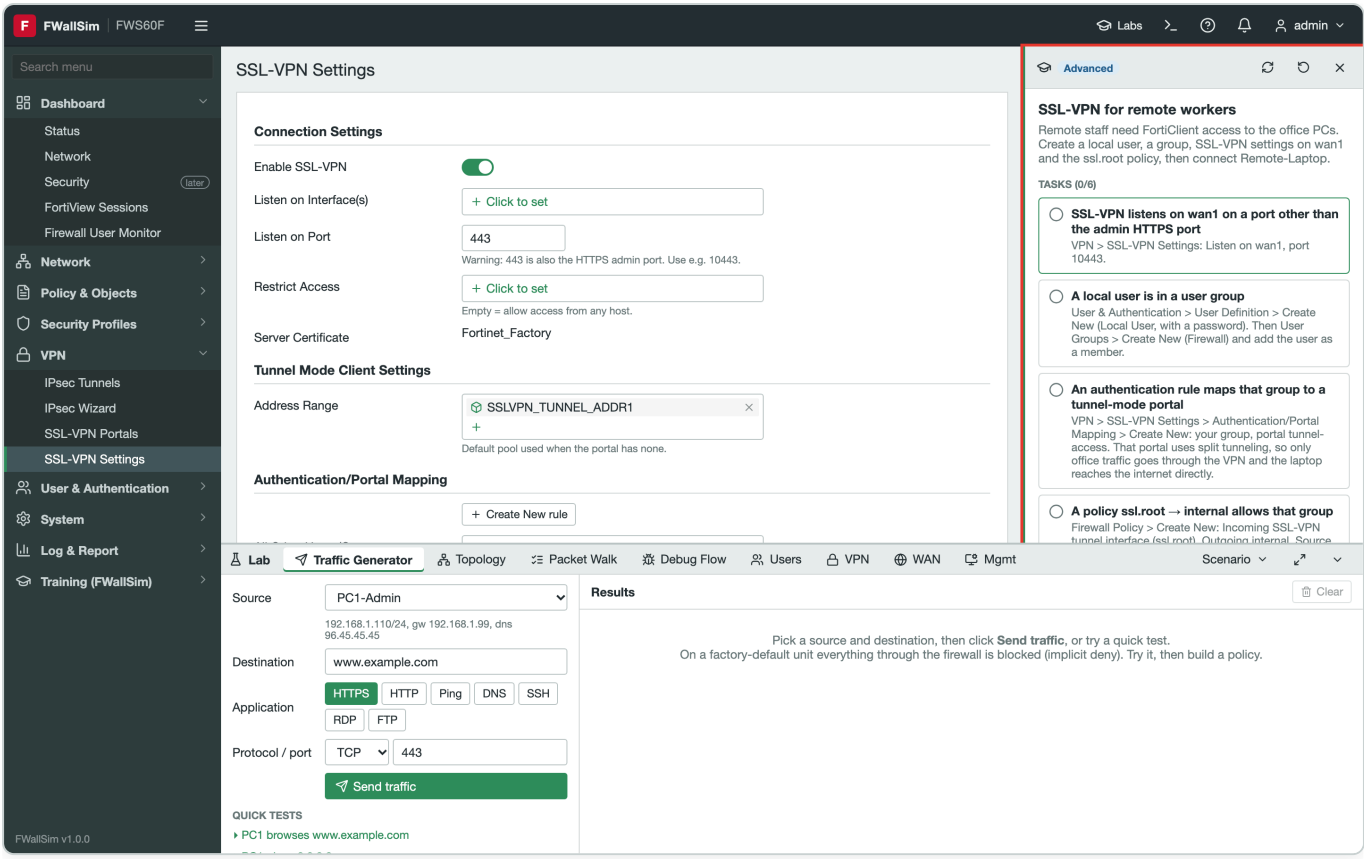
   Lab > VPN: connect Remote-Laptop to 203.0.113.10 on your SSL-VPN port with the local user.

6. **Remote-Laptop can RDP to PC1 through the tunnel**

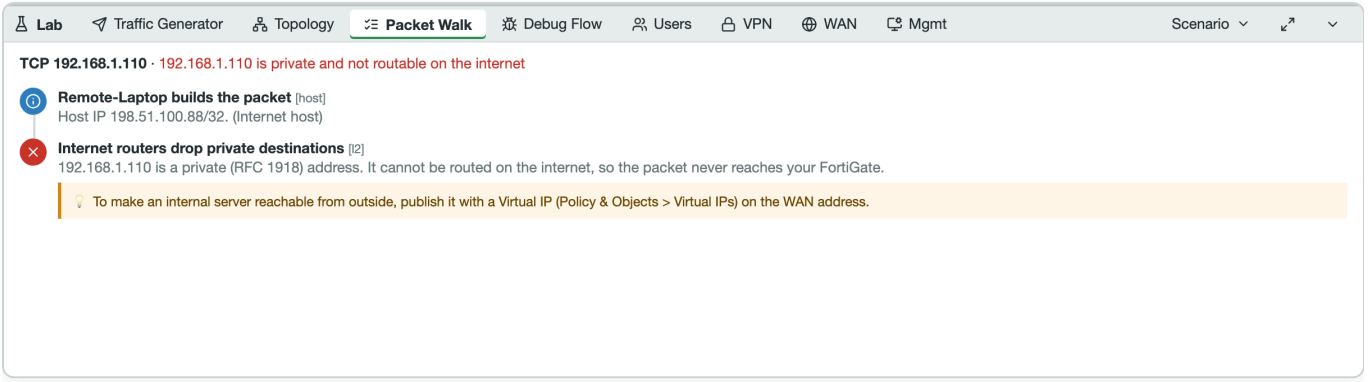
   Send Remote-Laptop → 192.168.1.110 TCP/3389.

Screenshots

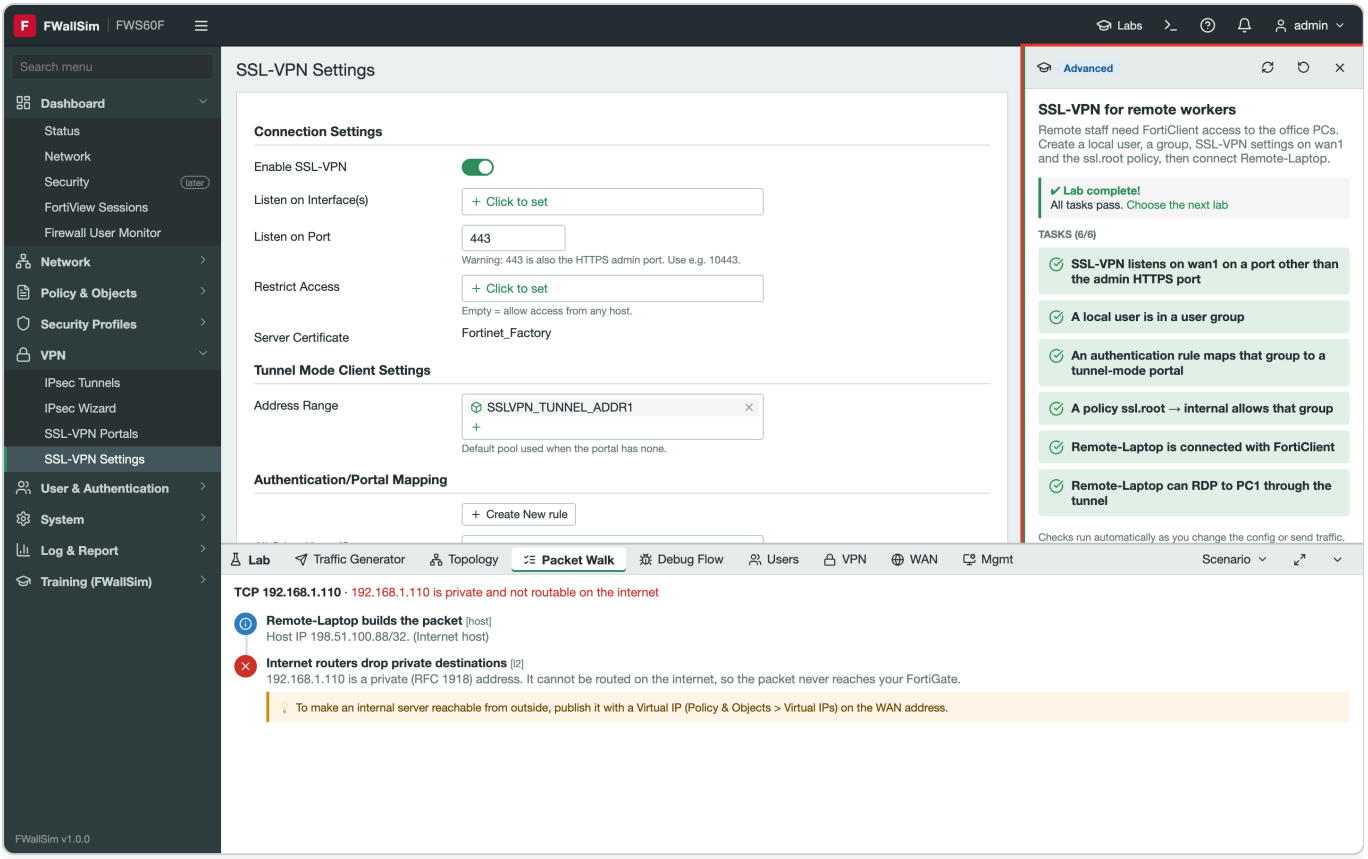
Numbered red markers show where to click or what to look at.



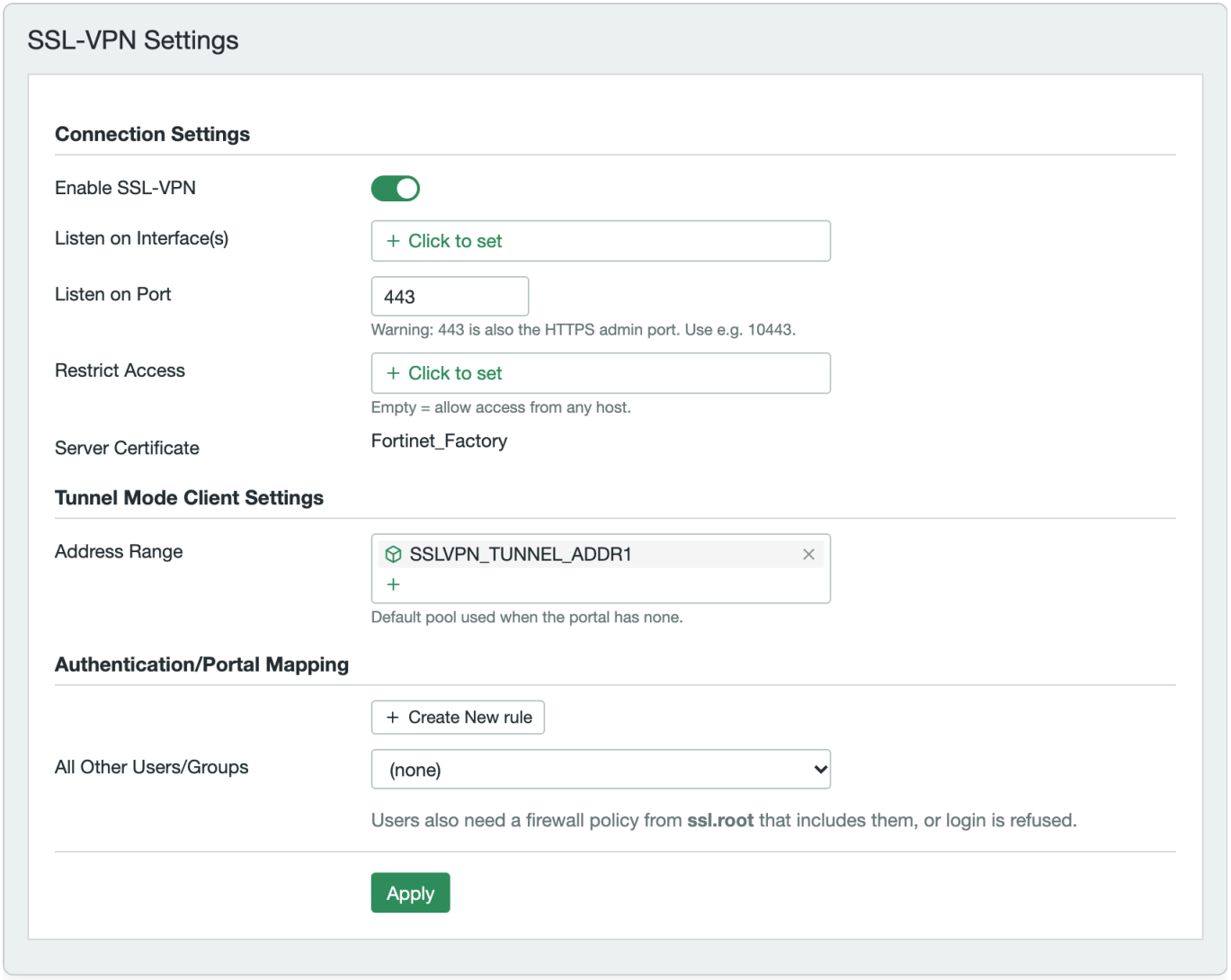
Lab Started



Packet Walk Before


FWallSim | FWS60F
Labs
admin
Search menu
Dashboard
Status
Network
Security
later
FortiView Sessions
Firewall User Monitor
Network
Policy & Objects
Security Profiles
VPN
IPsec Tunnels
IPsec Wizard
SSL-VPN Portals
SSL-VPN Settings
User & Authentication
System
Log & Report
Training (FWallSim)
FWallSim v1.0.0
SSL-VPN Settings
Connection Settings
Enable SSL-VPN
Listen on Interface(s)
+ Click to set
Listen on Port
443
Warning: 443 is also the HTTPS admin port. Use e.g. 10443.
Restrict Access
+ Click to set
Empty = allow access from any host.
Server Certificate
Fortinet_Factory
Tunnel Mode Client Settings
Address Range
SSLVPN_TUNNEL_ADDR1
Default pool used when the portal has none.
Authentication/Portal Mapping
+ Create New rule
Advanced
SSL-VPN for remote workers
Remote staff need FortiClient access to the office PCs. Create a local user, a group, SSL-VPN settings on wan1 and the ssl.root policy, then connect Remote-Laptop.
✔ Lab complete!
All tasks pass. Choose the next lab
TASKS (6/6)
SSL-VPN listens on wan1 on a port other than the admin HTTPS port
A local user is in a user group
An authentication rule maps that group to a tunnel-mode portal
A policy ssl.root → internal allows that group
Remote-Laptop is connected with FortiClient
Remote-Laptop can RDP to PC1 through the tunnel
Checks run automatically as you change the config or send traffic.
Lab
Traffic Generator
Topology
Packet Walk
Debug Flow
Users
VPN
WAN
Mgmt
Scenario
TCP 192.168.1.110 · 192.168.1.110 is private and not routable on the internet
Remote-Laptop builds the packet [host]
Host IP 198.51.100.88/32. (Internet host)
Internet routers drop private destinations [l2]
192.168.1.110 is a private (RFC 1918) address. It cannot be routed on the internet, so the packet never reaches your FortiGate.
To make an internal server reachable from outside, publish it with a Virtual IP (Policy & Objects > Virtual IPs) on the WAN address.

Lab Solved



SSL-VPN Settings

Firewall Policy

+ Create New | Edit | Delete | Policy Lookup | Interface Pair View | By Sequence

Search

Name	ID	From	To	Source	Destination	Schedule	Service	Actic
internal → wan1 (1)								
LAN-to-Internet	1	internal	wan1	LAN-Subnet	all	always	ALL	✓ A(
wan1 → dmz (1)								
Web-Server-Inbound	2	wan1	dmz	all	VIP-WebServer-HTTP	always	HTTP	✓ A(
dmz → wan1 (1)								
DMZ-to-Internet	3	dmz	wan1	WebServer-DMZ	all	always	DNS HTTP HTTPS	✓ A(
ssl.root → internal (1)								
ssl	4	ssl.root	internal	all VPN	LANnet	always	ALL	✓ A(
Implicit (1)								
Implicit Deny	0	any	any	all	all	always	ALL	⊘ D

Policies are evaluated from top to bottom and the **first match wins**. Drag rows to reorder them. Right-click for more actions (insert, clone, enable/disable, logs). Right-click the Implicit Deny row to log violation traffic.

Firewall Policy

Lab | Traffic Generator | Topology | Packet Walk | Debug Flow | Users | VPN | WAN | Mgmt | Scenario

TCP 192.168.1.110 · Remote-Laptop → PC1-Admin TCP/3389: delivered and answered, allowed by policy 4

Remote-Laptop builds the packet [host]
Host IP 198.51.100.88/32. (Internet host)

FortiClient sends it through the SSL-VPN tunnel [l2]
Inner packet 10.212.134.200 → 192.168.1.110 (split tunneling decided this destination goes into the tunnel). Outer packet: TLS from 198.51.100.88 to the FortiGate. It appears on interface ssl.root.

Ingress: packet arrives on ssl.root [ingress]
TCP 10.212.134.200:56381 → 192.168.1.110:3389

Session lookup: new session 00002a01 [session]
No existing session matches this connection (5-tuple), so this is the first packet. It goes through the full checks: DNAT, routing, then policy. Later packets of the session take the fast path.

Destination NAT: none [dnat]
No Virtual IP matches this destination.

Route lookup: out via internal [route]
Connected route 192.168.1.0/24 on internal. This is the longest-prefix active route for 192.168.1.110, so the egress interface is internal.

Packet Walk After

Next: Class 12: SD-WAN with two ISPs